

An extension of the general criterion for importance of pore diffusion with chemical reactions

K. B. BISCHOFF

Department of Chemical Engineering, The University of Texas, Austin, Texas

(Received 4 April 1966; in revised form 24 October 1966)

Abstract—The general criterion for the importance of pore diffusion with reaction in porous solids as developed by Weisz and Prater is one of the most useful practical results of the effectiveness factor concept. This criterion uses only observable quantities and so can often be easily estimated. However, it was developed for only first order reactions and does not give proper results for other reaction types. Using recent work on generalized effectiveness factors, the criterion is here extended to include any reaction rate form.

ONE OF the most useful aspects of the effectiveness factor concept for reaction in porous solids is its use to define conditions such that pore diffusion is *not* important. For practical applications, particularly in development work, the exact details of any diffusion limitations are usually not desired beyond this assurance. WEISZ and PRATER [1] developed a criterion for this purpose such that the (usually unknown) reaction rate constant need not be used and *only* observable quantities were required. Their derivation, however, was for first order reactions only and PETERSEN [2] has recently pointed out that for other cases, particularly adsorption mechanisms, the details can be rather drastically different.

Recently, ARIS [3], BISCHOFF [4], and PETERSEN [5] have shown how the entire effectiveness factor concept can be generalized for any type of reaction rate form such that all curves are asymptotically approximately the same. With this basis, it would seem that a more general criterion of the Weisz-Prater type could be devised. PETERSEN [2] has shown how this can be done for the special case of an adsorption mechanism and partially for a general reaction rate form. However, his final formula still contains the reaction rate constant, and so does not use *only* directly observable quantities.

To see how these two concepts may be combined, a brief recapitulation of each of them will first be given.

1. GENERAL EFFECTIVENESS FACTOR

ARIS [3], BISCHOFF [4], and PETERSEN [5]† have all shown that a “generalized” or “normalized” modulus can be defined which makes the effectiveness factor curves for all reaction rate forms asymptotically follow essentially the *same* curve.

The modulus for any type of reaction is,

$$(mL) = \frac{Lr(C_o)}{\sqrt{2}} \left[\int_{C_{eq}}^{C_o} D_e(c)r(c)dc \right]^{-\frac{1}{2}} \quad (1)$$

where C_{eq} is the equilibrium concentration that would be attained at the center of an infinite pellet. The characteristic length, L , is here written as for a flat plate, but ARIS [7] has shown that when properly interpreted, it is the same for other shaped particles as well. Also, it should be mentioned that for a general single reaction, where the rate may be a function of all compositions, the rate function can still be written in terms of only one composition variable, as in Eq. (1); this is shown, for example, by PETERSEN [2]. Variable diffusivity, volume changes on reaction, and other complications can also be handled in Eq. [1]. An example of the use for reversible reactions is for a simple first order case with constant diffusivity:

† An early treatment with many of the same features was by STEWART [6].

$$r(C) = \frac{k}{K}[(1+K)C - A] \quad (2)$$

$$C_e = \frac{A}{1+K}$$

where

A = sum of surface concentrations of reactant and product.

Substitution of Eq. (2) into Eq. (1) followed by some algebraic manipulations gives,

$$(mL) = L \sqrt{\left(\frac{k}{D_e} \frac{1+K}{K}\right)} \quad (3)$$

Equation (3) was found by CARBERRY [8] to bring all the curves for different values of the equilibrium constant, K , asymptotically together, as should be true for the general modulus.

The use of the general modulus, Eq. (1), for non-isothermal reactions has not been considered in detail previously. PETERSEN [9] presented some asymptotic formulas which contain the elements of the general treatment, and ARIS [10] showed how Eq. (1) could in principle be used. With the usual Arrhenius form of rate constant, it can be shown [5, 9, 11, 12] that the particle temperature can be eliminated in terms of concentration to give a rate form

$$r(C) = k_{vo} C \exp \left[\gamma \beta \frac{C_o - C}{(1 + \beta)C_o - \beta C} \right] \quad (4)$$

With the substitution

$$y \equiv \frac{\gamma C_o}{(1 + \beta)C_o - \beta C}$$

the integral in Eq. (1) can be found with Eq. (4), giving (similar to that found by PETERSEN [9]),

$$(mL) = L \sqrt{\left(\frac{k_{vo}}{D_e}\right) \frac{\beta}{\sqrt{(2\gamma)e^{\gamma/2}}}} \left\{ -\left(\beta + \frac{1}{2} + \frac{\gamma}{2}\right) \frac{e^{-\gamma}}{\gamma} + \left[\frac{(\beta + 1)^2 + \gamma(\beta + 1)}{2} \right] \frac{e^{-(\gamma/1 + \beta)}}{\gamma} \right\}$$

$$-\left(\beta + 1 + \frac{\gamma}{2}\right) \left[E_1\left(\frac{\gamma}{1 + \beta}\right) - E_1(\gamma) \right] \quad (5)$$

where

$$E_1(\alpha) = \int_{\alpha}^{\infty} \frac{e^{-y}}{y} dy$$

= exponential integral.

The use of this modulus would then asymptotically bring together the curves for both exothermic and endothermic reactions. The endothermic curves tend to follow the other isothermal cases rather closely, but because of the phenomenon of having an effectiveness factor greater than unity for exothermic reactions, these portions stay separated. Figure 1 shows a selection of isothermal curves for single order† and adsorption mechanisms from BISCHOFF [4] along with non-isothermal cases from CARBERRY [13], WEEKMAN [14], and WEISZ and HICKS [12].

Due to the complexity of Eq. (4), it is often approximated by a simpler exponential form, good for $\beta \ll 1$:

$$r(C) = k_{vo} C e^{\gamma \beta} e^{-\gamma \beta (C/C_o)} \quad (6)$$

The use of Eq. (6) gives a general modulus

$$(mL) = L \sqrt{\left(\frac{k_{vo}}{D_e}\right) \frac{\gamma \beta}{\sqrt{[2(e^{\gamma \beta} - 1 - \gamma \beta)]}}} \quad (7)$$

which would be simpler to use when applicable.

Thus, from the previous references and the present discussion of reversible and non-isothermal reactions, the general modulus, Eq. (1), can be used with confidence for any reaction type.

† HUGO [15] has shown that the curves hold for any order $0 < n \rightarrow \infty$.

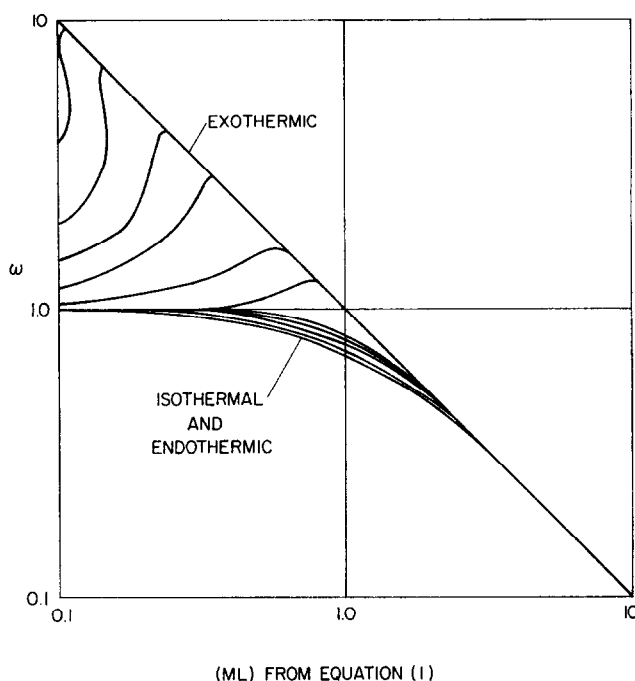


FIG. 1. Plot of effectiveness factors, ε , for all reaction types using generalized modulus.

2. WEISZ-PRATER CRITERION

This discussion of the WEISZ-PRATER [1] derivation will indicate the manipulations required to obtain the criterion.

The observed, directly measurable rate is,

$$r_{\text{obs}} = \varepsilon r(c_{\text{obs}}) \quad (8)$$

where $r(c_{\text{obs}})$ indicates the reaction rate functional form evaluated at the *bulk* observed concentration, c_{obs} . The effectiveness factor, ε , is a function of the modulus, which for a first order reaction is,

$$mL = L\sqrt{(k_v/D_e)} \quad (9)$$

The rate constant, k_v , is usually unknown and so Weisz and Prater algebraically eliminated it for the criterion.

If Eq. (8) is specialized to a first order reaction, it is seen how to eliminate k_v :

$$r_{\text{obs}} = k_v c_{\text{obs}} \varepsilon(mL) \quad (10) \quad \text{which is the criterion for a first order reaction.}$$

$$= \frac{D_e c_{\text{obs}}}{L^2} (mL)^2 \varepsilon(mL) \quad (11)$$

where k_v is substituted from Eq. (9) to give Eq. (11). This step is crucial and can be done easily here since Eq. (9) is explicitly solvable for k_v .

If all the directly observable and independently measurable variables are put on one side of the equation, we obtain,

$$\frac{r_{\text{obs}} L^2}{c_{\text{obs}} D_e} = (mL)^2 \varepsilon(mL) \quad (12)$$

The Weisz-Prater argument is now based on the general shape of the effectiveness factor plot for first order reactions shown in Fig. 1. It is seen that for $mL < \sim 1.0$ that $\varepsilon = 1.0$ (no diffusion limitations) so that in this region, even without knowledge of the exact value of (mL) , Eq. (12) says that,

$$\frac{r_{\text{obs}} L^2}{c_{\text{obs}} D_e} < 1 \quad (13)$$

3. GENERAL CRITERION

To formulate a general criterion, one should combine the general modulus, Eq. (1), with Eq. (8), analogous to the Weisz-Prater derivation. In other words, the rate constant should be solved for from Eq. (1) and substituted into Eq. (8). In the absence of a definite functional form, however, an explicit solution is not possible, and so this is where PETERSEN [2] stopped.

It seems that one further step is possible in that most reaction rate forms used in practice can be written,

$$r(c) = k_v g(c) \quad (14)$$

where k_v is the "main" rate constant about which little is known in the type of studies discussed above. There may be other parameters in $g(c)$ but these often can at least be approximately estimated independently of k_v . For example, some forms of $g(c)$ could be,

$$g(c) = c^n$$

$$g(c) = \frac{c}{1 + Kc}$$

Thus it is seen that for isothermal reactions, a wide variety of cases could be considered. For non-isothermal situations, the temperature variable part of the rate constant must also be included in $g(c)$ [as in Eq. (4)] and so the activation energy must be estimated beforehand. This can often be approximately done, but does limit the generality of the method for these cases.

With only this limitation to the general reaction rate functional form, the problem can be solved to obtain a criterion. If Eq. (14) is substituted into Eq. (1), one obtains,

$$(mL) = \sqrt{k_v \frac{Lg(C_{\text{obs}})}{\sqrt{2}}} \left\{ \int_{C_{\text{eq}}}^{C_{\text{obs}}} D_e(c)g(c)dc \right\}^{-\frac{1}{2}} \quad (15)$$

The unknown rate constant, k_v , can now be explicitly found from Eq. (15) and substituted into Eq. (8), and after some rearrangement it is found that,

$$\frac{r_{\text{obs}} L^2 g(C_{\text{obs}})}{2 \int_{C_{\text{eq}}}^{C_{\text{obs}}} D_e(c)g(c)dc} = (mL)^2 \varepsilon (mL) \quad (16)$$

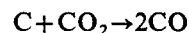
Figure 1 shows that the effectiveness factor is always significantly less than unity only when the modulus is greater than one, just as for the first order case treated by WEISZ and PRATER [1]. This is true for *all* reaction forms of the figure, and so the general criterion would then be, by analogy with the steps in going from Eq. (12) to Eq. (13),

$$\boxed{\frac{r_{\text{obs}} L^2 g(c_{\text{obs}})}{2 \int_{C_{\text{eq}}}^{C_{\text{obs}}} D_e(c)g(c)dc} < 1} \quad (17)$$

Equation (17) should then be of general applicability for a wide range of reaction types and gives a general definite criterion *not* limited to first order reactions.

For the case of exothermic reactions, the criterion does not show when $\varepsilon = 1.0$ but it does exclude the region where $\varepsilon < 1.0$. For practical purposes, one is usually most interested in whether or not pore diffusion effects will cause the reaction rate and thus conversion to be lower than expected. If the rate is higher, $\varepsilon > 1.0$, there would not be a problem, except for such things as harming the catalyst by too high a temperature. WEI [11] has presented methods for telling when $\varepsilon \rightarrow 1.0$, but they are fairly complicated. Thus, for the special case of exothermic reactions the general criterion, Eq. (17), will still indicate the region where pore diffusion will not lower the reaction rate, $\varepsilon \geq 1.0$, but will do so conservatively in the sense that the rate could be higher than expected, but not lower.

The use of the general criterion, Eq. (17), will next be illustrated by applying it to a situation where the first order criterion, Eq. (13), gives completely erroneous results. PETERSEN [2] used some data of AUSTIN and WALKER [16] on the carbon-carbon dioxide reaction,



to show how the first order criterion would not be correct; the data will here be re-calculated on the basis of Eq. (17). This reaction seems to be very strongly inhibited by adsorption of the product, carbon monoxide, which leads to large deviations from first order behavior. The rate equation, in concentration units, was of the standard adsorption type,

$$r = \frac{k_v C}{1 + K_2 C' + K_3 C} \quad (18)$$

where

C = concentration of CO_2

C' = concentration of CO

k_v = rate constant

K_2, K_3 = adsorption constants

From the reaction stoichiometry and assuming equal diffusivities and zero carbon monoxide concentration at the particle surface, Eq. (18) becomes

$$r = \frac{k_v C}{(1 + 2K_2 C_o) + (K_3 - 2K_2)C} \quad (19)$$

where

$C_o = C_{\text{obs}}$ = concentration of CO_2 at particle surface.

If Eq. (19) is substituted into the general criterion, Eq. (17), one obtains,

$$\frac{r_{\text{obs}} L^2}{2D_e} \left\{ \frac{1 + K_3 C_o}{K_3 - 2K_2} \left[1 - \frac{1 + 2K_2 C_o}{C_o (K_3 - 2K_2)} \right] \right. \\ \left. \ln \frac{1 + K_3 C_o}{1 + 2K_2 C_o} \right\}^{-1} < 1 \quad (20)$$

At 1000°K , the following data were used (re-calculated on concentration basis):

$$C_o(1 \text{ atm}) = 1.22 \times 10^{-5} \text{ mole/cm}^3$$

$$K_2 = 4.15 \times 10^9 \text{ cm}^3/\text{mole}$$

$$K_3 = 3.38 \times 10^5 \text{ cm}^3/\text{mole}$$

$$D_e = 0.1 \text{ cm}^2/\text{sec}$$

$$L = 0.7 \text{ cm}$$

$$r_{\text{obs}} = 5.67 \times 10^{-9} \text{ mole/cm}^3\text{-sec}$$

Substitution of these values into the Weisz-Prater first order criterion, Eq. (13), gives,

$$\frac{r_o L^2}{C_o D_e} = 2.28 \times 10^{-3} < 1.0$$

Thus the criterion is satisfied (< 1.0) and would indicate that pore diffusion limitations did not exist. However, by cutting apart the particles and observing the profiles of reacted carbon and from other tests, Austin and Walker found that there were indeed large diffusion effects present, and so the first order criterion did not correctly predict the behavior.

If the data are now substituted into the general criterion, Eq. (20), one obtains,

$$\text{LHS(Eq. (20))} = 2.5 > 1.0$$

Since the value of Eq. (20) is *greater* than one, the criterion is *not* satisfied and so pore diffusion effects should be present, as found by Austin and Walker. Therefore, the general criterion, Eq. (17), indicated the proper situation and so should prove useful for similar tests for any reaction type.

NOTATION

A	sum of reactant and product concentrations at particle surface
C	concentration
C_{eq}	equilibrium reactant concentration
C_o	particle surface concentration
$C_{\text{obs}} = C_o$	observed concentration at surface of particle
D_e	effective diffusivity of particle

E activation energy	L size of particle
ε effectiveness factor	m modulus
g rate functional form without rate constant	$r(c)$ reaction rate functional form
$(-\Delta H)$ heat of reaction	r_{obs} observed reaction rate at surface of particle
k_e effective thermal conductivity of particle	T_o surface temperature of particle
k_v reaction rate constant	y integration variable
k_{vo} reaction rate constant at particle surface	β parameter $= (-\Delta H)D_e C_o / T_o k_e$
	ν parameter $= E / RT_o$

REFERENCES

- [1] WEISZ P. B. and PRATER C. D., *Adv. Catalysis* 1954 **6** 167.
- [2] PETERSEN E. E., *Chem. Engng Sci.* 1965 **20** 587.
- [3] ARIS R., *Ind. & Engng Chem. Fundls* 1965 **4** 227; 1965 **4** 487.
- [4] BISCHOFF K. B., *Am. Inst. Chem. Engrs JI* 1965 **11** 351.
- [5] PETERSEN E. E., *Chemical Reaction Analysis*, Prentice Hall, New York 1965.
- [6] STEWART W. E., in *Notes on Transport Phenomena*, BIRD R. B., STEWART W. E. and LIGHTFOOT E. N., p. 335, Wiley, New York 1958.
- [7] ARIS R., *Chem. Engng Sci.* 1957 **6** 262.
- [8] CARBERRY J. J., *Am. Inst. Chem. Engrs JI* 1962 **8** 557.
- [9] PETERSEN E. E., *Chem. Engng Sci.* 1962 **17** 987.
- [10] ARIS R., *Introduction to the Analysis of Chemical Reactors*, Prentice Hall, New Jersey 1965.
- [11] WEI J., *Chem. Engng Sci.* 1965 **20** 729.
- [12] WEISZ P. B. and HICKS J. S., *Chem. Engng Sci.* 1962 **17** 265.
- [13] CARBERRY J. J., *A.I.Ch.E.JI* 1961 **7** 350.
- [14] WEEKMAN V. W., *J. Catalysis* 1966 **5** 44.
- [15] HUGO P., private communication.
- [16] AUSTIN L. G. and WALKER P. L., *Am. Inst. Chem. Engng JI* 1963 **9** 303.

Résumé—Le critère général de l'importance de la diffusion à travers les pores, avec réaction dans des solides poreux, développé par Weisz et Prater, est l'un des résultats pratiques les plus utiles du concept du facteur d'efficacité. Ce critère utilise seulement les quantités observables et se laisse donc souvent facilement évaluer. Toutefois, il a seulement été développé pour des réactions de premier ordre et ne donne pas de bons résultats pour les autres types de réactions. Utilisant des travaux récents sur les facteurs généralisés d'efficacité, le critère est ici étendu pour inclure toutes les formes de taux de réaction.

Zusammenfassung—Der durch Weisz und Prater entwickelte Massstab für die Beurteilung der Bedeutung der Porendiffusion mit Reaktion in porösen Feststoffen stellt eines der brauchbarsten Ergebnisse des Wirksamkeitsfaktorbegriffes dar. Diese Beurteilung stützt sich nur auf beobachtbare Grössen und sie kann daher oft auch einfache Weise eingeschätzt werden. Der Massstab ist jedoch nur für Reaktionen erster Ordnung ermittelt worden und gibt für andere Reaktionstypen keine brauchbaren Resultate. Auf Grund neuerer Arbeiten über verallgemeinerte Wirksamkeitsfaktoren, ist die Anwendbarkeit des Beurteilungsmassstabes hier auf beliebige Reaktionsformen erweitert worden.